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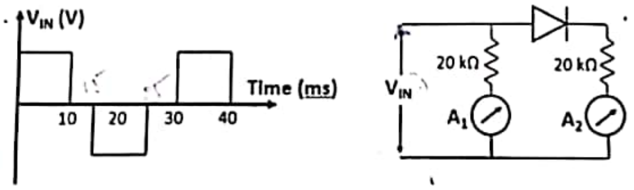
ELECTRONICS ENGINEERING DEPARTMENT IIT (ISM) DHANBAD
SEMESTER MONSSON 2018-1019

EXAMINATION: END SEMESTER EXAMINATION
SEMESTER B.TECH. 1ST SEMESTER (GROUP E, F, G, H) — TIME: 3 HRS
SUBJECT: ECC 11101 ELECTRONIC ENGINEERING — MAX MARKS: 100

INSTRUCTIONS: ANSWER ALL QUESTIONS FROM PART I, AND ANY TWO QUESTIONS FROM PART II.
MAKE SUITABLE ASSUMPTIONS WHEREVER NECESSARY

PART I

Q. 1	State whether the following statements are TRUE or FALSE (T/F). Give appropriate reasoning (R) for your answer in brief. a. An ideal op amp has voltage gain of the order of 10^6 A/A. b. Waveforms can be shaped easily by using a combination of diode, capacitor and battery. c. A rectifier converts alternating current into a unidirectional current. In a half-wave rectification with input voltage $V_m \sin\omega t$ and a resistive load R_L, the dc current $I_{dc} = V_m/\pi R$. d. In a digital circuit we have the output given as $\overline{(\bar{A} + \bar{B}) + \bar{C}}$. This expression is equivalent to BC + AC. e. The number $(5F)_{16}$ is equivalent to $(96)_{10}$.	5X(1(T/F)+3(R)) = 20 Marks
Q.2 (a)	The signal V_{IN} is applied to the circuit shown in Figure 1. Predict the reading on the two meters A_1 and A_2, if they are DC ammeters.  Figure 1	10 marks
Q.2 (b)	A four-variable logic function comprises of following max-terms: $\prod M$ (3,7,9,13,15). Deduce the minimized SOP form using k-map. Draw the corresponding logic circuit.	10 marks
Q3 (a)	A Silicon semiconductor doped with 10^{17} cm^{-3} phosphorus atoms has been used to design a humidity sensor. In May 2018, the sensor was tested at Delhi, India (35 °C); Aziziyah, Libya (50 °C); Melbourne, Australia (14 °C) and Talvik, Norway (-5 °C). Numbers inside the parenthesis represent the temperature of the place. When an electric field of 1 MV/m was applied, 16 kA/cm² current density was observed in the Silicon sensor. Assuming the diffusivity to be independent of temperature, in which place/city the measurement was performed? ~~Electron mobility 0.135 m²/V.S, Hole mobility 0.048 m²/V.S~~	10 marks

$D_n = 2.18\ cm^2/s$, $D_p = 1\ cm^2/s$

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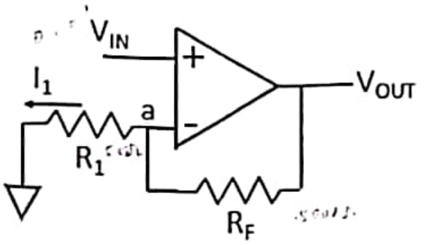
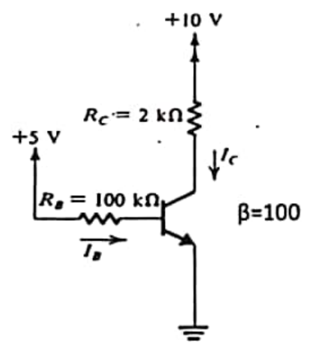
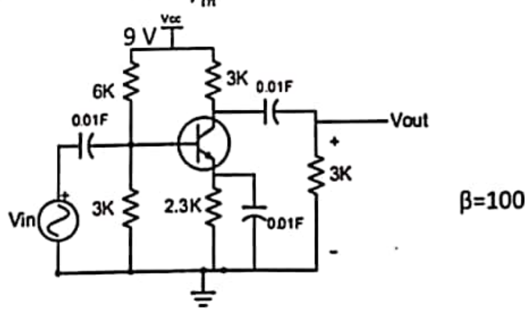
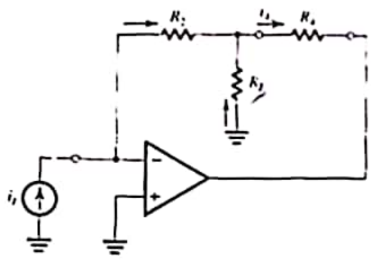
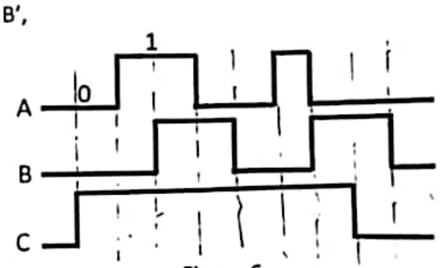
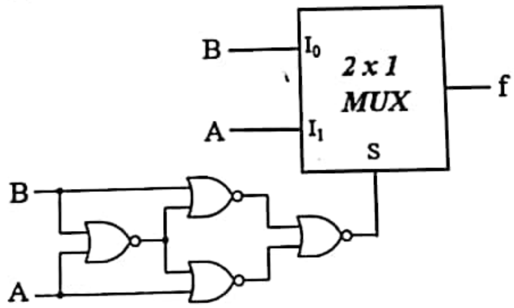
Q. 3 (b)	An ideal op-amp amplifier is shown in Figure 2, where R_1 = 5 kΩ and R_F = 500 kΩ. Evaluate the voltage gain V_{OUT}/V_{IN}. For an input voltage V_{IN} = 0.01 V, what is V_a and I_1?  Figure 2	10 marks
	PART II	
Q.4 (a)	For the circuit shown in Figure 3, determine the currents I_B, I_C, I_E and V_{CE}. Assume V_{BE}= 0.7 V, V_T= 25 mV.  Figure 3	10 marks
Q. 4 (b)	A CE amplifier circuit is shown in Figure 4. Assume V_{BE}= 0.7 V, V_T= 25 mV. i. Find the DC collector current (I_C) and the collector to emitter voltage (V_{CE}). ii. Draw the small signal equivalent circuit. iii. Evaluate the mid-band voltage gain $\frac{V_{out}}{V_{in}}$  Figure 4	10 marks

Q5 (a)	Design an op-amp adder circuit to give an input-output relationship of: $V_{out} = 4V_1 + 5V_2$, where V_1 and V_2 are voltages applied toward the inverting and non-inverting terminal respectively. Assume the value of feedback resistor as 100 KΩ.	10 marks
Q. 5 (b)	For the circuit shown in Figure 5, show that $i_4 = i_1 (1+\frac{R_2}{R_3})$. Assume the op amp to be ideal.  Figure 5	10 marks
Q6 (a)	Using the following waveforms shown in Figure 6, draw the waveform of X, where If X = AC + B',  Figure 6	10 marks
Q6 (b)	Using the above wavefunctions A and B, draw the waveform of f for the logic circuit in Figure 7..  Figure 7	10 marks

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